

Supplementary Material for *Hierarchical Knowledge Distillation for Fair Federated Learning*

1 Overview

This document contains supplementary material for the paper “*Hierarchical Knowledge Distillation for Fair Federated Learning*”.

2 Additional Details of the Datasets

The experiments were conducted using three widely recognized benchmarks datasets in Federated Learning (FL): MNIST, CIFAR-10, and FEMNIST¹. The MNIST dataset consists of grayscale images of handwritten digits (28×28 pixels) across 10 classes. In contrast, CIFAR-10 includes 32×32 pixel RGB images of natural scenes, also classified into 10 object categories. Both of these datasets are originally centralized and IID (Independent and Identically Distributed). To introduce statistical heterogeneity, the training data was partitioned across clients using Dirichlet sampling [2] with a concentration parameter of $\alpha = 1$. This method creates uneven label distributions and varying dataset sizes among the clients. FEMNIST, which is derived from the extended MNIST dataset, contains handwritten characters from 62 classes. This dataset is naturally partitioned by writer, resulting in inherently non-IID client datasets.

Table 1: Client dataset statistics after partitioning. The columns show the number of classes (Cls), the number of clients (Clt), the total training samples (Total), and the average (Avg), minimum (Min), and maximum (Max) number of samples per client.

Dataset	Cls	Clt	Total	Avg	Min	Max
MNIST	10	92	32,666	355	40	1,330
CIFAR-10	10	127	47,371	373	52	1,668
FEMNIST	62	3,574	733,419	205	78	525

The dataset statistics (see Table 1) indicate significant variation in client dataset sizes, particularly for MNIST and CIFAR-10 under Dirichlet partitioning, reflecting realistic heterogeneity in federated environments.

¹ Dataset links: MNIST: <http://yann.lecun.com/exdb/mnist>, CIFAR-10: <https://www.cs.toronto.edu/~kriz/cifar.html>, FEMNIST: <https://leaf.cmu.edu>

3 Implementation Details

In this section, we outline the model architectures, hyperparameters, training schedules, and other implementation details necessary for ensuring reproducibility, which were omitted from the main paper due to space constraints.

3.1 Model Architectures

MNIST: We use a lightweight convolutional architecture consisting of two convolutional layers (32 and 64 channels, kernel size 3×3 , padding 1) with ReLU activations, followed by 2×2 max pooling and dropout ($p = 0.25$). The convolutional features are flattened and passed through a fully connected layer with 128 units before the final 10-class classifier.

CIFAR-10. The model consists of two convolutional blocks. Each block contains two 3×3 convolution layers with group normalization and ReLU activations, followed by 2×2 max pooling. The convolutional backbone is followed by a fully connected layer with 256 units and dropout ($p = 0.3$), and a final 10-class output layer.

FEMNIST: We employ a convolutional architecture with three convolutional blocks. Each block consists of a 3×3 convolution layer followed by group normalization and ReLU activation, with 2×2 max pooling for spatial downsampling. The resulting feature map is flattened and passed through a fully connected layer with 256 units and dropout ($p = 0.3$) before the final classifier with 62 classes.

3.2 Choice of Hyperparameters.

For all algorithms (FEDAVG [3], Q-FEDAVG [1], and DEFFT), we maintained consistent local and global training budgets: 5 local epochs and 300 global rounds for MNIST, and 10 local epochs over 300 global rounds for CIFAR-10 due to its complexity, and 5 local epochs over 150 global rounds for FEMNIST. Optimization is performed using mini-batch stochastic gradient descent (SGD). The learning rate is set to 0.01 for FEMNIST and 0.001 for both MNIST and CIFAR-10. The training batch size is 64 for FEMNIST and 32 for MNIST and CIFAR-10. For Q-FEDAVG, the fairness parameter q was chosen empirically: $q = 0.5$ for MNIST and $q = 0.4$ for CIFAR-10 and FEMNIST. Higher q values did not consistently enhance tail performance and could reduce stability.

3.3 Training Configuration

All experiments were conducted in a simulated FL environment implemented in Python, with model training performed using the PyTorch framework. The experiments were executed on a single device equipped with four NVIDIA H200 NVL GPUs (CUDA 13.1), an AMD EPYC 9565 processor (2 sockets, 72 cores per socket; 288 logical CPUs), and 1 TiB of system memory, where multiple clients are simulated sequentially during each communication round. To ensure reproducibility, the complete implementation and experimental scripts are publicly available at: Anonymous GitHub Repository.

4 Additional Experimental Results

4.1 Client Clustering Visualization

To illustrate the heterogeneity of client data distributions, we visualize the client label-distribution representations used for clustering. Each client k is represented by the normalized class distribution vector

$$d_k = [d_{k1}, d_{k2}, \dots, d_{kC}],$$

where d_{kc} denotes the proportion of samples belonging to class c . For visualization, these vectors are projected to two dimensions using t-SNE, and colors correspond to the cluster assignments obtained by the clustering procedure described in Section 5 of the submitted manuscript.

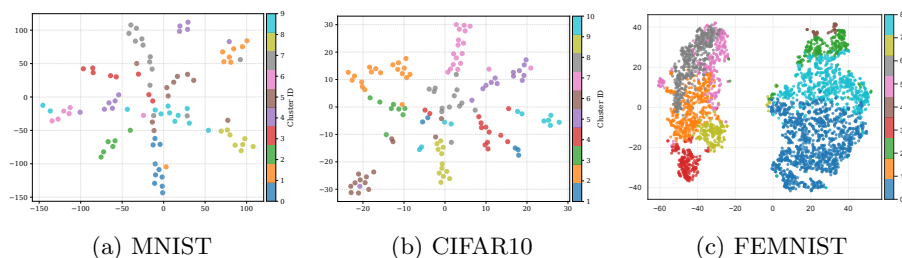


Fig. 1: Visualization of client label-distribution embeddings using t-SNE. The x-axis and y-axis correspond to the first and second dimensions of the t-SNE embedding (t-SNE Dimension 1 and t-SNE Dimension 2), respectively. Each point represents a client and is colored according to its assigned cluster. The embeddings are derived from the normalized class-distribution vectors of clients. Distinct clusters indicate groups of clients with similar label distributions, which motivates the hierarchical aggregation strategy used in DEFFT.

As shown in Fig. 1, clients form clearly separated groups in the embedding space, indicating substantial label-distribution heterogeneity. Specifically, the clustering procedure identifies 10 clusters for MNIST, 11 clusters for CIFAR-10, and 9 clusters for FEMNIST. MNIST and CIFAR-10 exhibit moderate cluster separation due to the synthetic Dirichlet partitioning, whereas FEMNIST shows stronger structure reflecting its naturally heterogeneous writer-based data distribution. These clusters motivate the hierarchical organization of clients used in DEFFT.

4.2 Training Dynamics Across Communication Rounds

Fig. 2 shows the change in global test accuracy over communication rounds for MNIST Ctrl St, MNIST, CIFAR-10, and FEMNIST. Results are averaged over

five independent runs with different random client participation. Across datasets, DEFFT consistently demonstrates faster early-stage convergence compared to FEDAVG and Q-FEDAVG. The improvement is most noticeable in the MNIST Ctrl St, where stronger client heterogeneity leads to a clear accuracy advantage throughout training. On the standard MNIST, DEFFT achieves similar final accuracy while converging more rapidly in the early communication rounds. For CIFAR-10 and FEMNIST, all methods converge to comparable final performance, indicating that DEFFT maintains competitive performance even in more complex or naturally distributed settings.

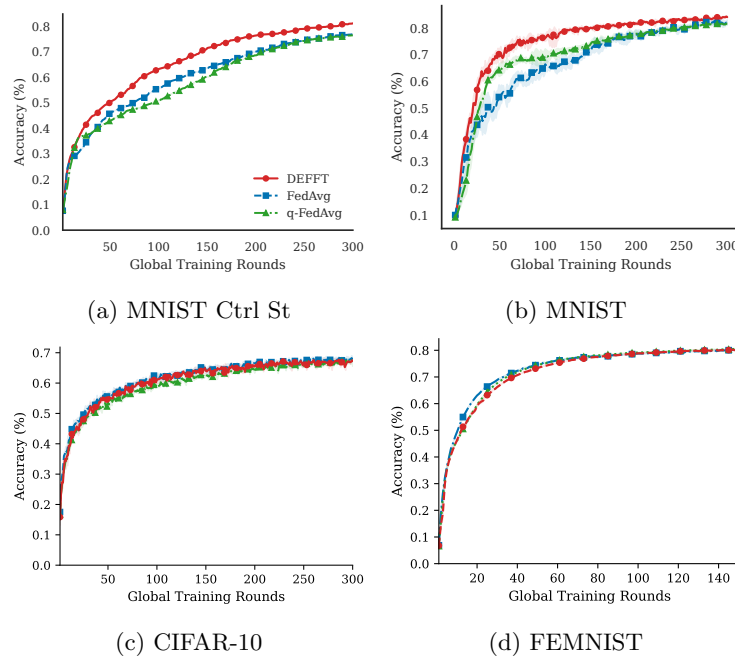


Fig. 2: Results are averaged over five independent runs with different random client participation.

References

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2. Lin, T., et al.: Ensemble distillation for robust model fusion in federated learning. *Advances in neural information processing systems* **33**, 2351–2363 (2020)
3. McMahan, B., et al.: Communication-efficient learning of deep networks from decentralized data. In: *AI and Statistics*. pp. 1273–1282. PMLR (2017)